

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

AI LAB

Assignment-10

Name: Ansik Singh Tomar

Roll No: 2401037

Branch: CSE

Course: Artificial Intelligence Lab

Title: Tic-Tac-Toe Game using Minimax Algorithm

1. Objective

To implement Tic-Tac-Toe game using Minimax algorithm where the AI plays optimally.

2. Description

The Tic-Tac-Toe game is a two-player game played on a 3x3 grid. One player is human and the other is AI. The AI uses the Minimax algorithm to choose the best possible move.

3. Algorithm

Minimax recursively evaluates all possible game states. It assigns scores based on win, loss, or draw. The AI selects the move that maximizes its chances of winning.

4. Code Explanation

The program includes functions to print the board, check the winner, implement minimax, and determine the best move.

5. Sample Input and Output

User selects X or O and plays by entering row and column. AI responds with optimal moves.

6. Result

The AI always plays optimally and never loses.

7. Conclusion

The Minimax algorithm ensures optimal gameplay and demonstrates decision-making in AI.